

Paper 2, Theorem 1.3(iv): Exponent-Domain Amendment

Internal research note for the **ShenWork** formalization

13 July 2026

Abstract

The proof of Theorem 1.3(iv) in Chen–Ruau–Shen invokes the weighted elliptic estimate of Proposition 2.2 at an exponent which is not necessarily above one. The missing admissibility condition is $q_* > 2 - 2m$, where $q_* = \max\{1, N\alpha/2\}$. This note derives the condition, gives an explicit admitted parameter wedge where the printed argument fails, and records the corrected theorem proved in Lean. The issue is a proof gap; no counterexample to the boundedness conclusion is claimed.

1 Source and printed alternative

The source examined is Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1, 16 December 2025. The local archival copy is **paper2.pdf**.

Theorem 1.3(iv) assumes

$$\beta \geq \frac{1}{2}, \quad \alpha = 2m + \gamma - 2, \tag{1}$$

together with the smallness alternative (1.25). It is stated for all positive m, γ, α .

2 The missing exponent-domain condition

Section 5.4 invokes Proposition 2.2 at

$$s(P) = \frac{P + \alpha}{\gamma}. \tag{2}$$

Proposition 2.2 requires $s(P) > 1$. Substituting (1) gives

$$\begin{aligned} \frac{P + \alpha}{\gamma} > 1 &\iff P + 2m + \gamma - 2 > \gamma \\ &\iff P > 2 - 2m. \end{aligned}$$

The proof selects a seed exponent immediately to the right of

$$q_* := \max \left\{ 1, \frac{N\alpha}{2} \right\}. \tag{3}$$

It therefore needs

$$\boxed{q_* > 2 - 2m}, \tag{4}$$

or equivalently $(q_* + \alpha)/\gamma > 1$. Condition (4) is not implied by the printed hypotheses. The claim in Section 5.4 that the exponent supplied to Proposition 2.2 is above one for all seed powers is false in the remaining wedge.

3 Concrete admitted parameter wedge

On the interval choose

$$N = 1, \quad m = \frac{1}{4}, \quad \gamma = \frac{7}{2}, \quad \alpha = 2, \quad \beta = 1. \quad (5)$$

Then $\alpha = 2m + \gamma - 2$, while

$$q_* = \max\{1, \alpha/2\} = 1, \quad s(q_*) = \frac{1+2}{7/2} = \frac{6}{7} < 1. \quad (6)$$

Furthermore $(N\alpha - 2)_+ = 0$. The first disjunct in (1.25) is therefore automatic, so the printed theorem places no smallness condition on χ_0 . Proposition 2.2 becomes available only for $P > 3/2$. At such powers the squared-chemotaxis coefficient is nonzero, and the automatic first disjunct of (1.25) provides no estimate producing the required low seed.

This demonstrates a gap in the written proof, not a counterexample to the boundedness statement. A new estimate may recover part or all of the uncovered parameters.

4 Proof-supported amendment

The Section 5.4 route is valid after adding (4). On the one-dimensional interval this reads

$$\max\{1, \alpha/2\} > 2 - 2m. \quad (7)$$

The literal profile defining (1.19) is then continuous at q_* . The smallness condition (1.25) yields $p_0 > q_*$ with strict damping. Since $p_0 > \alpha/2$, a one-dimensional fixed-target interpolation absorbs $\int u^{P+\alpha}$ at one finite target P above both m and γ . A restarted Neumann-semigroup estimate then gives the supremum bound. No infinite Moser iteration is needed.

Recommended amendment to alternative (iv). In addition to the printed hypotheses, assume $\max\{1, N\alpha/2\} > 2 - 2m$. Then the proof in Section 5.4 is valid. The original statement outside this range remains unproved by the written argument.

5 Lean realization

The formalization keeps the missing condition explicit.

- `IntervalDomainTheorem13CriticalConstants.lean` realizes the literal M^* , q_* , and right-sided lower limit K .
- `IntervalDomainTheorem13CriticalSeed.lean` proves the case-(iv) energy estimate with $2 - 2m < P$.
- `IntervalDomainTheorem13CriticalThreshold.lean` obtains the low seed from (1.25) under $q_* > 2 - 2m$.
- `IntervalDomainTheorem13CriticalBootstrap.lean` proves `boundedBefore_critical_case_iv_corrected` on the faithful general- m interval model by a single fixed-target interpolation and the proved restarted endpoint.

All exported capstones are `sorry`-free and depend only on `propext`, `Classical.choice`, and `Quot.sound`.

Alternative (iii) does not share this defect: under $\alpha = m + \gamma - 1$, the inequality $(q_* + \alpha)/\gamma > 1$ follows from $q_* \geq 1$ and $m > 0$.

Reference

L. Chen, I. Ruau, and W. Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1 (2025), <https://arxiv.org/abs/2512.14858>.